Leaf and Non-Leaf Node Counting in a General Tree

Algorithm

Step 1: Start.

Step 2: Define a node structure containing data, a pointer to the first child, and a pointer to the next sibling.

Step 3: Initialize the root of the tree as NULL, and set leafCount = 0 and nonLeafCount = 0.

Step 4: Construct the general tree:

- Read the root value and create the root node.
- For each node, read the number of children.
- Create each child node and link it to its parent (first child, then next siblings).
- Repeat for every child until all nodes are created.

Step 5: Traverse the tree starting from the root (Preorder Traversal):

- Visit the current node.
- If the node has no child (first child = NULL), it is a leaf node: increase leafCount and store its data in the leaf list.
- Otherwise, it is a non-leaf node: increase nonLeafCount and store its data in the non-leaf list.
- Repeat the same for the first child and then for each of its siblings.

Step 6: Display all the leaf nodes and the value of leafCount.

Step 7: Display all the non-leaf nodes and the value of nonLeafCount.

Step 8: Display the total number of nodes (leafCount + nonLeafCount).

Step 9: Stop.